Supplementary Material for "Symmetry-Selected Nonlinear Thermal Pumping in Periodically Heated Mixed-Convection Channels"

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This document collects the derivations, the verification records and the supporting figures and tables of the study. It is referenced from the main manuscript throughout: every item below is cited there, and every number in this document carries an **S** (Fig. S1, Table S1, Sec. S1, Eq. (S1)) so that the two documents can never be confused when they cite each other. Numbers quoted without an **S** are those of the main manuscript, whose reference list is the only one: this document adds no citation of its own.

S1 Perturbative parameter maps and scaling checks

Table S1: Fixed-Pe dependence of $A = \Pi_T/Gr$ at $k = 0.8$, $Gr = 100$.

Pe	$Re = 5$	$Re = 10$	$Re = 20$	$Re = 35$
2	-4.5382×10^{-4}	-3.9078×10^{-5}	6.1165×10^{-5}	4.5851×10^{-5}
4	-1.2126×10^{-3}	-3.1884×10^{-4}	-1.3565×10^{-5}	2.4576×10^{-5}
7	-1.5951×10^{-3}	-4.7485×10^{-4}	-6.4279×10^{-5}	6.7452×10^{-6}
12	-1.6138×10^{-3}	-5.0178×10^{-4}	-8.5084×10^{-5}	-4.0374×10^{-6}
20	-1.5130×10^{-3}	-5.0141×10^{-4}	-1.0914×10^{-4}	-1.9812×10^{-5}

Table S2: Long-wavelength fit at $(Re, Pr, Gr) = (10, 0.7, 100)$.

exponent p	prefactor	R^2	standard error	number of points
1.9955	2.1665	1.0000	0.00133	7

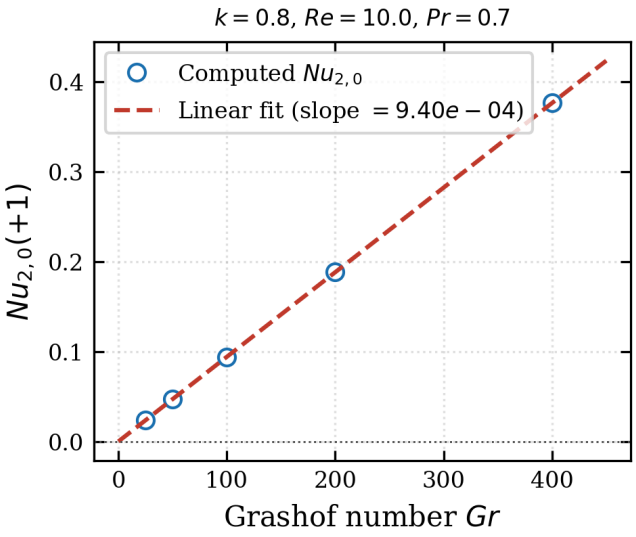


Figure S1: Thermal pumping coefficient versus Grashof number at fixed $(k, Re, Pr) = (0.8, 10, 0.7)$. The free-intercept fit is effectively exact and the response vanishes in the forced-convection limit $Gr = 0$.

Table S3: Test of Corollary 1 at $k = 0.8$, $Pe = 7$. The normalized response is invariant to a worst-case relative spread of 7.3×10^{-11} and the reversal Reynolds number is identical to the nine displayed digits for all four Grashof numbers (absolute spread 1.2×10^{-10}); the scan is archived as group B of the data package.

Gr	$A(Re = 25)$	$A(Re = 30)$	$A(Re = 35)$	Re_c
25	-2.089882×10^{-5}	-1.958845×10^{-6}	$+6.745165 \times 10^{-6}$	30.838496520
50	-2.089882×10^{-5}	-1.958845×10^{-6}	$+6.745165 \times 10^{-6}$	30.838496520
100	-2.089882×10^{-5}	-1.958845×10^{-6}	$+6.745165 \times 10^{-6}$	30.838496520
200	-2.089882×10^{-5}	-1.958845×10^{-6}	$+6.745165 \times 10^{-6}$	30.838496520

S2 Verification of the nonlinear DNS

S3 Phase decomposition, term balance and transport comparison

S4 Two-dimensional physical structure of the mechanism

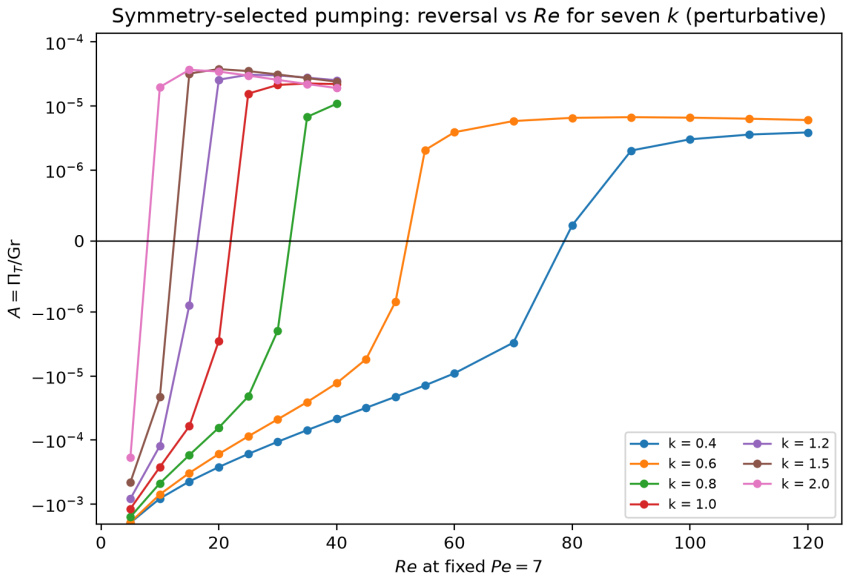


Figure S2: Normalized pumping response $A = \Pi_T/Gr$ versus Reynolds number at fixed $Pe = 7$ and $Gr = 100$ for seven wavenumbers, computed with the authoritative perturbative solver. Each curve crosses zero exactly once. The $k = 0.4$ and $k = 0.6$ curves are drawn over the extended interval $Re \leq 120$ that their crossings require; the remaining five use the primary $Re \leq 40$ window, inside which each of them already crosses. The crossing moves to lower Re as k increases.

Table S4: Independent nonlinear DNS at fixed $Pe = 7$, $k = 0.8$, and $Gr = 100$. The Prandtl number is adjusted according to $Pr = Pe/Re$.

Re	Pr	A_{DNS}	A_{pert}	Relative difference
5	1.4000	-1.5878×10^{-3}	-1.5951×10^{-3}	0.46%
10	0.7000	-4.6763×10^{-4}	-4.7485×10^{-4}	1.52%
20	0.3500	-5.9977×10^{-5}	-6.4279×10^{-5}	6.69%
30	0.2333	$+7.5042 \times 10^{-7}$	–	–
35	0.2000	$+8.9842 \times 10^{-6}$	$+6.7452 \times 10^{-6}$	33.19%

Table S5: Term-wise balance of Π_T at $k = 0.8$, $Pe = 7$, $Gr = 100$ with $Pr = Pe/Re$. The two dominant contributions keep fixed signs—the streamwise advection term is positive, the wall-normal sweep term negative—and the reversal occurs when the faster decay of the sweep term lets the two cancel; the term-by-term evaluation is archived as group B of the data package.

Re	$\mathcal{L}[\langle U^{(a)}\partial_X\Theta^{(s)}\rangle]$	$\mathcal{L}[\langle V^{(a)}\partial_Y\Theta^{(s)}\rangle]$	$\mathcal{L}[\langle U^{(s)}\partial_X\Theta^{(a)}\rangle]$	$\mathcal{L}[\langle V^{(s)}\partial_Y\Theta^{(a)}\rangle]$	Π_T
5	5.504×10^{-2}	-1.870×10^{-1}	-2.537×10^{-2}	-2.236×10^{-3}	-1.595×10^{-1}
10	5.899×10^{-2}	-9.726×10^{-2}	-2.707×10^{-3}	-6.504×10^{-3}	-4.749×10^{-2}
20	3.737×10^{-2}	-4.251×10^{-2}	$+4.895 \times 10^{-3}$	-6.180×10^{-3}	-6.428×10^{-3}
30	2.378×10^{-2}	-2.420×10^{-2}	$+4.829 \times 10^{-3}$	-4.611×10^{-3}	-1.959×10^{-4}
Re_c	2.300×10^{-2}	-2.326×10^{-2}	$+4.761 \times 10^{-3}$	-4.494×10^{-3}	≈ 0
40	1.655×10^{-2}	-1.599×10^{-2}	$+3.939 \times 10^{-3}$	-3.420×10^{-3}	$+1.081 \times 10^{-3}$
50	1.232×10^{-2}	-1.154×10^{-2}	$+3.157 \times 10^{-3}$	-2.614×10^{-3}	$+1.330 \times 10^{-3}$

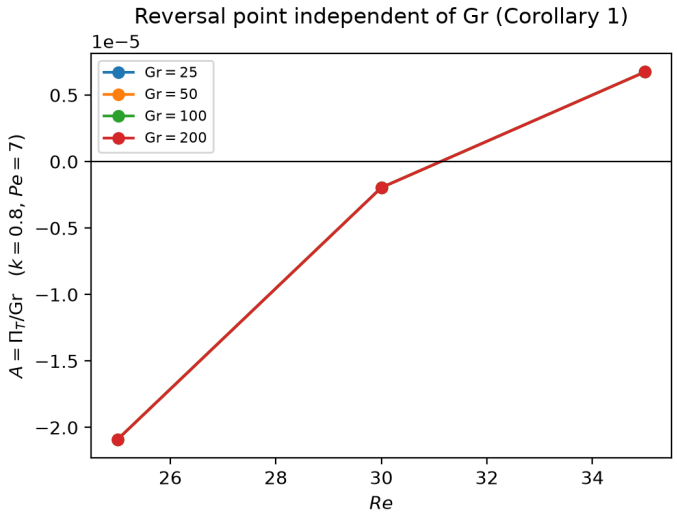


Figure S3: Normalized response $A = \Pi_T/Gr$ versus Re at $k = 0.8$ and $Pe = 7$ for four Grashof numbers. The four curves coincide, as required by Corollary 1.

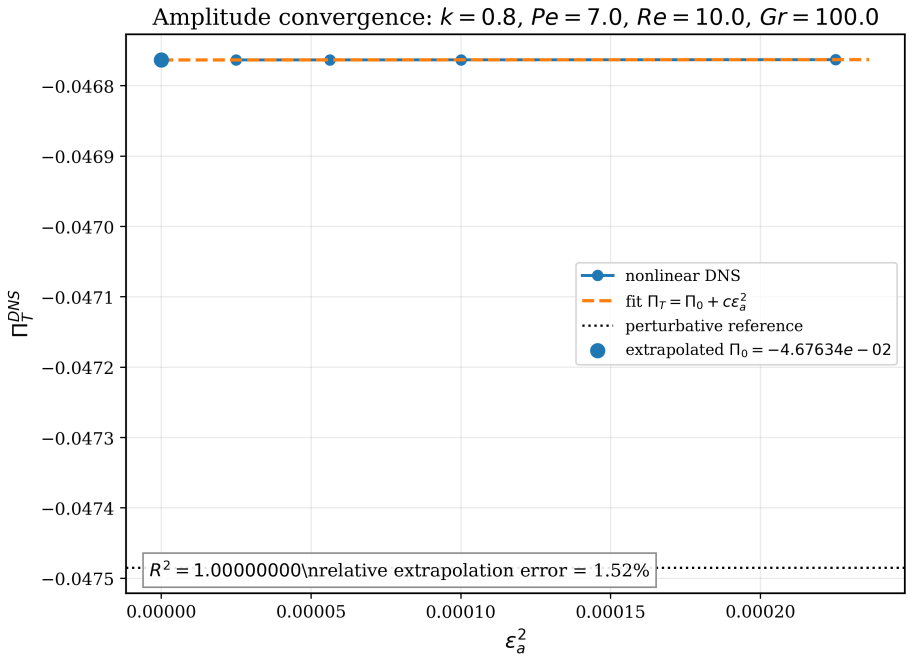


Figure S4: Convergence of the fully nonlinear DNS pumping coefficient with decreasing asymmetry amplitude. The nonlinear DNS values approach a finite $\varepsilon_a \to 0$ limit, which differs by only 1.52% from the second-order perturbative prediction. The result provides an independent nonlinear validation of the asymptotic pumping coefficient.

Table S6: Symmetry-based comparison of the leading nonlinear transport response.

Property	Symmetry-preserving forcing	Asymmetric forcing
Wall parity	even or purely odd	mixed s–a content
Leading mixed forcing	absent	odd $\mathcal{E}_{as}$
Mean mixed temperature	symmetry-preserving	odd in Y
Wall-flux difference	zero	generally nonzero
Leading consequence	no antisymmetric redistribution	transverse thermal pumping

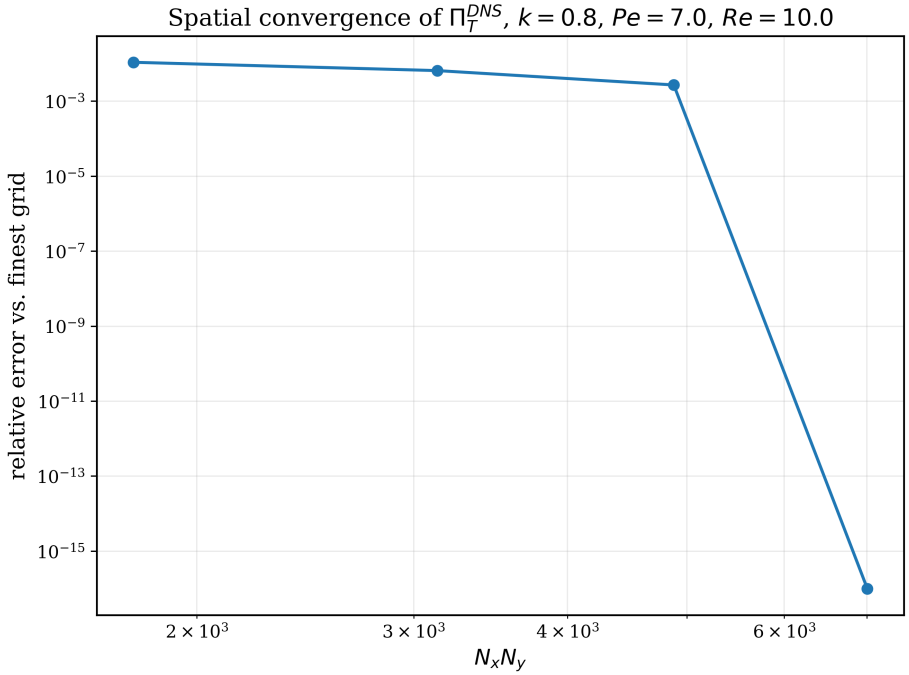


Figure S5: Spatial convergence of the nonlinear DNS pumping coefficient. The solution approaches a well-defined finest-grid value as both the streamwise and transverse resolutions are increased. The remaining difference between the finest-grid DNS result and the perturbative reference is smaller than the total discrepancy observed on coarser meshes.

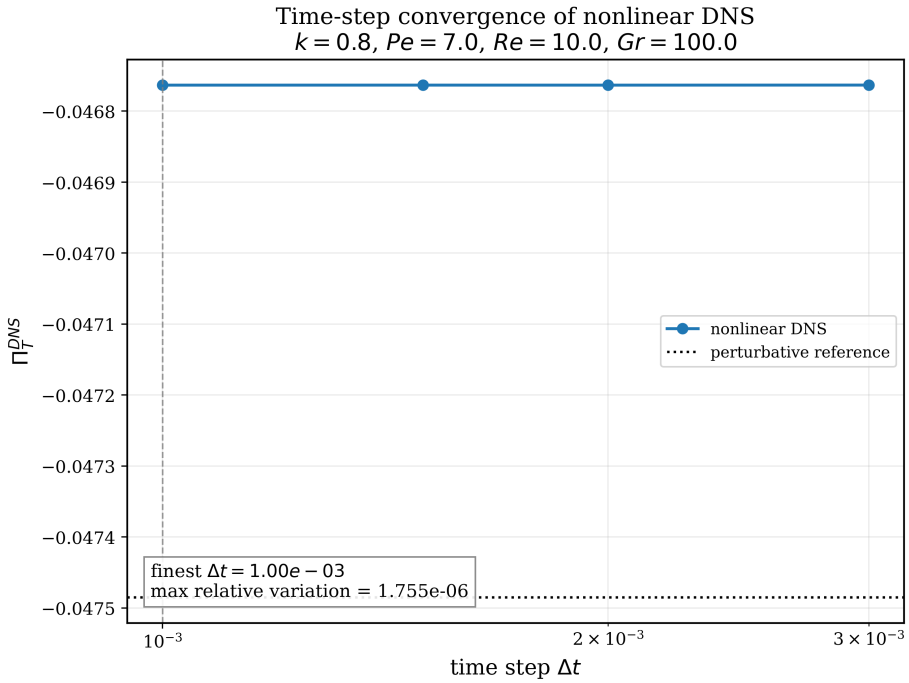


Figure S6: Time-step convergence of the nonlinear DNS pumping coefficient. The response remains unchanged to within approximately 10^{-6} in relative terms over the tested time-step range, confirming that the stationary wall-flux observable is temporally converged.

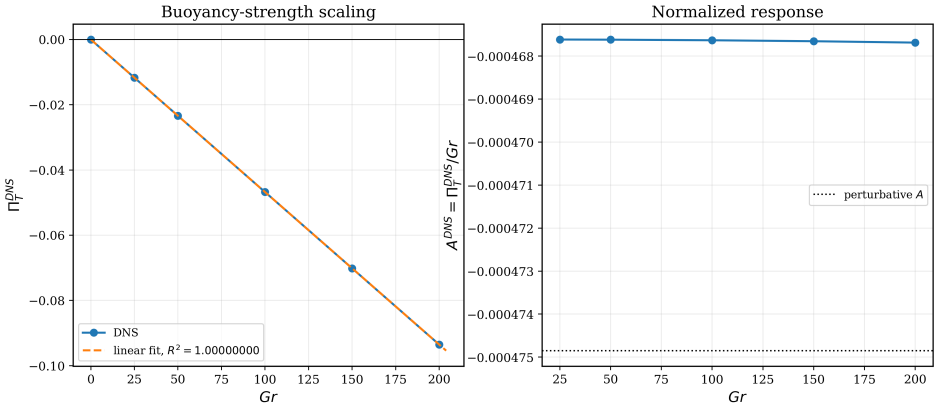


Figure S7: Nonlinear DNS verification of the Grashof-number scaling at fixed $(k, Re, Pr) = (0.8, 10, 0.7)$. The pumping coefficient is linear in Gr over the investigated interval, while the $Gr = 0$ calculation switches off the leading pumping response to numerical precision. The result independently confirms the analytical proportionality $\Pi_T = GrA(k, Re, Pr)$.

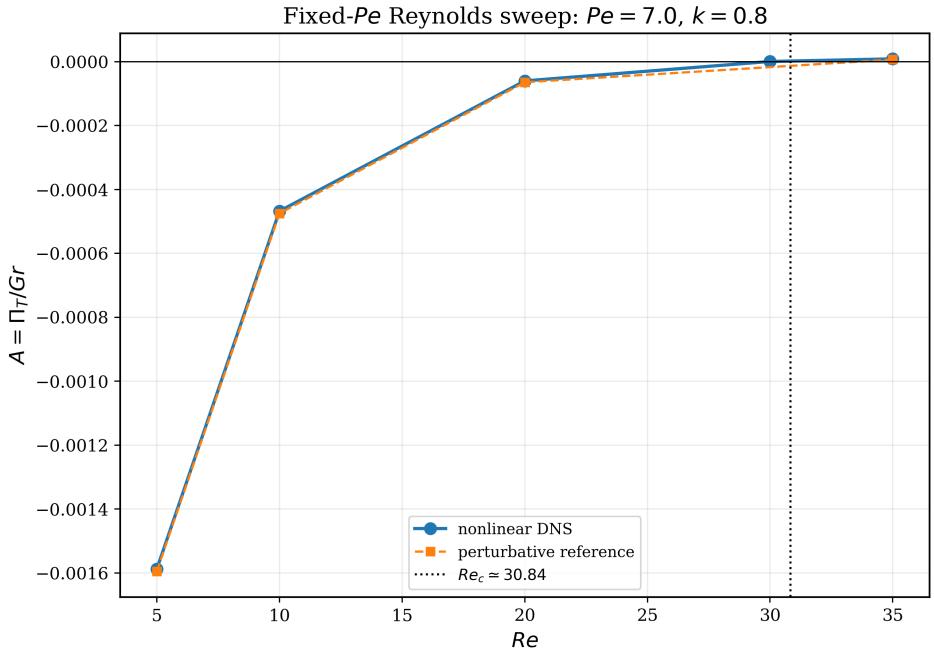


Figure S8: Normalized pumping response $A = \Pi_T/Gr$ from the independent nonlinear DNS at fixed $Pe = 7$, $k = 0.8$, and $Gr = 100$. Varying Re changes both the magnitude and the sign of the response. The transition from negative values at $Re = 20$ to positive values at $Re = 30$ and 35 provides direct nonlinear evidence that the hydrodynamic Reynolds number remains an active control parameter even when the Péclet number is held fixed.

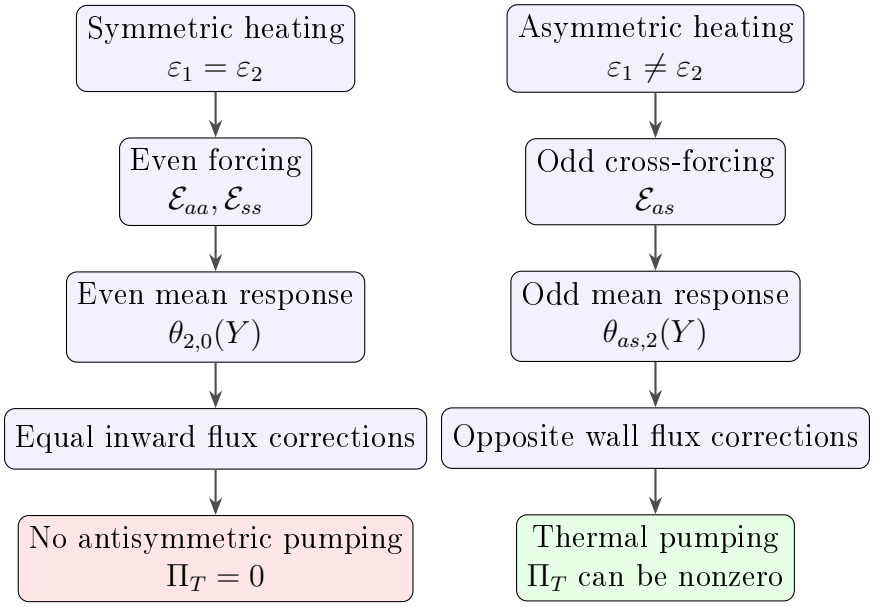


Figure S9: Logical chain connecting wall-temperature symmetry to the nonlinear wall-flux response.

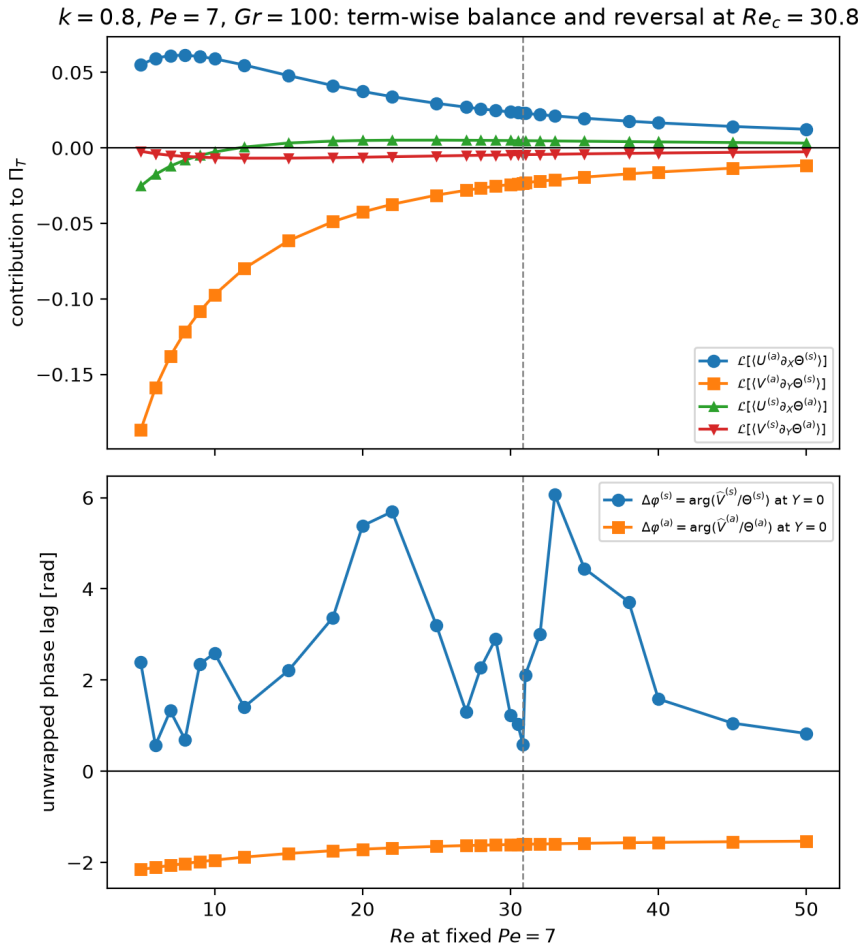


Figure S10: Upper panel: the four additive contributions (82) to Π_T versus Re at $k = 0.8$, $Pe = 7$; the dashed line marks $Re_c = 30.838$. Lower panel: unwrapped centreline phase lags, which vary smoothly and cross zero nowhere, showing that the reversal is a balance effect rather than a phase-sign effect.

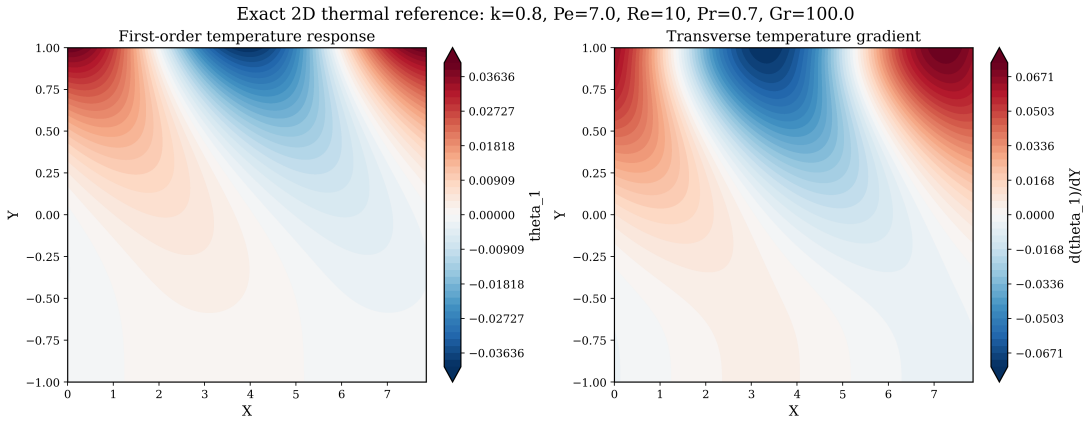


Figure S11: Two-dimensional thermal reference solution for $(k, Pe, Re, Pr, Gr) = (0.8, 7, 10, 0.7, 100)$. Left: first-order temperature response $\theta_1(X, Y)$. Right: transverse gradient $\partial_Y\theta_1$, which provides the leading buoyancy forcing of the hydrodynamic perturbation through Eq. (18).

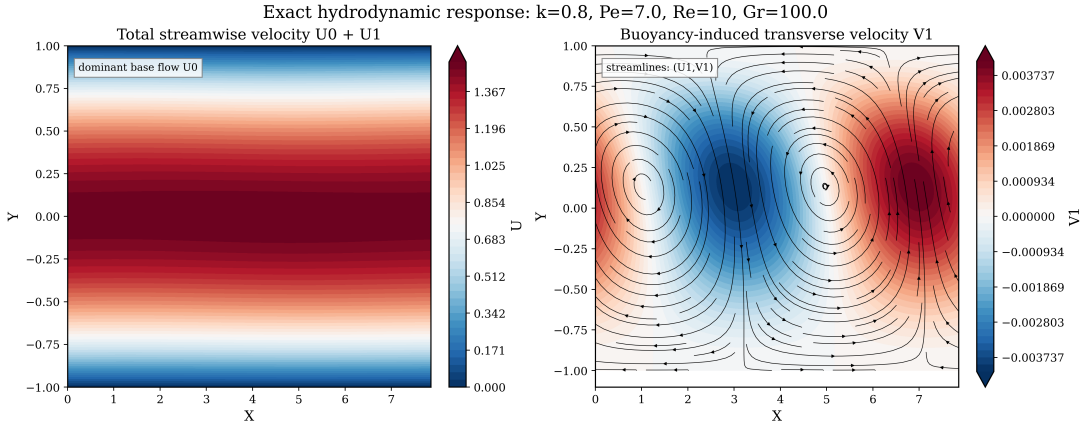


Figure S12: Hydrodynamic response for the reference case $(k, Pe, Re, Pr, Gr) = (0.8, 7, 10, 0.7, 100)$. Left: total streamwise velocity $U_0 + U_1$, showing the dominance of the Poiseuille base state. Right: buoyancy-induced transverse velocity V_1 with streamlines of the perturbation field (U_1, V_1). The secondary circulation, although small in amplitude, is the hydrodynamic carrier of the nonlinear thermal redistribution.

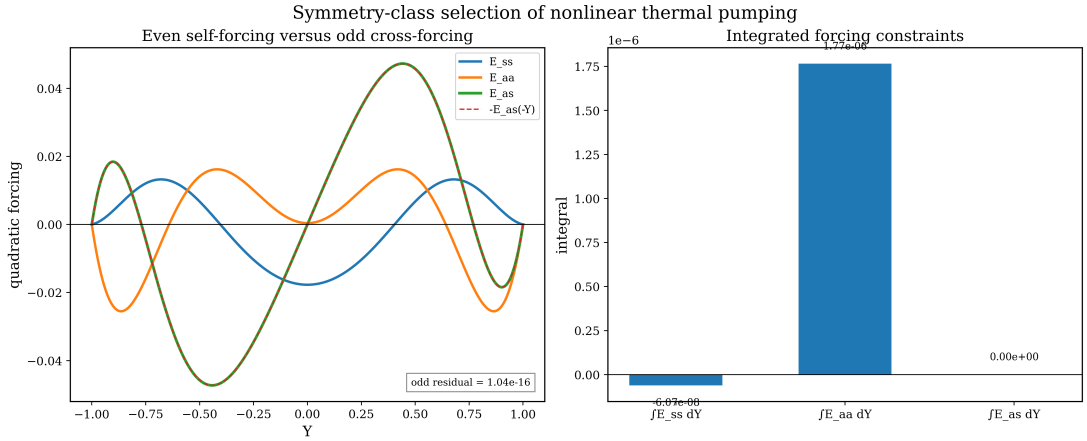


Figure S13: Symmetry classification of the quadratic thermal forcing for the reference configuration. Left: symmetric self-interactions $\mathcal{E}_{ss}$ and $\mathcal{E}_{aa}$ and mixed interaction $\mathcal{E}_{as}$. The latter satisfies $\mathcal{E}_{as}(-Y) = -\mathcal{E}_{as}(Y)$, with the displayed numerical parity residual of order 10^{-16}. Right: integrated forcing constraints. In particular, the mixed contribution has vanishing net integral, as required by its odd parity.

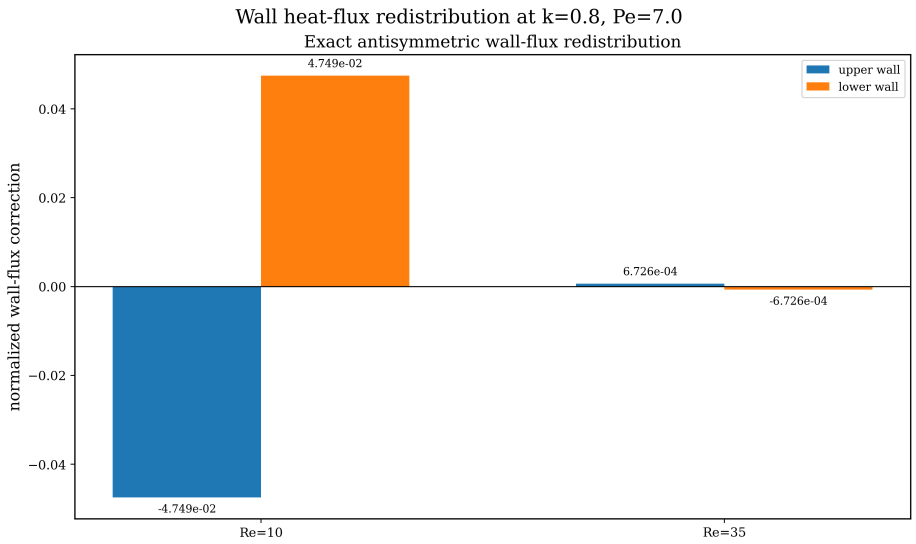


Figure S14: Mean nonlinear wall-flux redistribution for $k = 0.8$ and $Pe = 7$. The comparison is performed at fixed $Pe = 7$, with $Pr = Pe/Re$ adjusted accordingly. The upper and lower wall corrections are equal in magnitude and opposite in sign, in agreement with the exact relation $Nu_{2,0}(+1) = -Nu_{2,0}(-1)$. At $Re = 10$ the redistribution is comparatively strong and corresponds to $\Pi_T < 0$, whereas at $Re = 35$ the amplitude is much smaller and the sign has reversed.

S5 Derivations and independent checks

S5.1 Derivation of the first-order hydrodynamic BVP

For the e^{-ikX} convention, incompressibility gives $-ik\widehat{U}+\widehat{V}'=0$, hence $\widehat{U}=-i\widehat{V}'/k$. The complex perturbation vorticity is

$$\widehat{\omega}=\partial_X\widehat{V}-\partial_Y\widehat{U}=\frac{i}{k}(D^2-k^2)\widehat{V}. \tag{S1}$$

Taking the streamwise derivative of the transverse momentum equation minus the transverse derivative of the streamwise momentum equation yields

$$U_0\partial_X\omega_1-V_1U_0''=\frac{1}{Re}\nabla^2\omega_1-\frac{Gr}{Re^2}\partial_Y\theta_1. \tag{S2}$$

Substitution of the modal representation gives Eq. (18). The no-slip conditions imply $\widehat{V}(\pm1)=\widehat{V}'(\pm1)=0$.

S5.2 Reflection-operator proof

Define $\mathcal{R}[f](Y)=f(-Y)$. Since $\mathcal{R}D=-D\mathcal{R}$, $\mathcal{R}D^2=D^2\mathcal{R}$, and U_0, U_0'' are even,

$$\mathcal{R}\mathcal{L}_T=\mathcal{L}_T\mathcal{R}, \qquad \mathcal{R}\mathcal{L}_H=\mathcal{L}_H\mathcal{R}, \tag{S3}$$

where $\mathcal{L}_T$ and $\mathcal{L}_H$ denote the thermal and hydrodynamic operators. Uniqueness of the corresponding BVPs therefore fixes the parity classes of all first-order modal fields.

S5.3 Proof of the odd cross-interaction

Using Eq. (28), the four terms have parity

$$\widehat{U}^{(a)}\overline{-ik\Theta^{(s)}}:\text{odd}, \tag{S4}$$
$$\widehat{U}^{(s)}\overline{-ik\Theta^{(a)}}:\text{odd}, \tag{S5}$$
$$\widehat{V}^{(a)}\overline{(\Theta^{(s)})'}:\text{odd}, \tag{S6}$$
$$\widehat{V}^{(s)}\overline{(\Theta^{(a)})'}:\text{odd}. \tag{S7}$$

Therefore $\mathcal{E}_{as}$ is odd and its channel integral vanishes.

S5.4 Mean momentum balance and fixed-bulk-flow constraint

The mean second-order momentum equation is

$$\frac{1}{Re}U_{2,0}''+G_2=\langle U_1\partial_X U_1+V_1\partial_Y U_1\rangle_X-\frac{Gr}{Re^2}\theta_{2,0}. \tag{S8}$$

The conditions $U_{2,0}(\pm 1) = 0$ and $\int_{-1}^{1} U_{2,0}\, dY = 0$ determine G_2 under the fixed-bulk-flow formulation. This streaming problem is not required for the proof of wall-flux antisymmetry, but it is needed for the reported mean-flow validation.

S5.5 Proof of the nonlinear thermal-pumping theorem

From

$$\theta''_{as,2} = Pe\mathcal{E}_{as}, \qquad \theta_{as,2}(\pm 1) = 0, \tag{S9}$$

and oddness of $\mathcal{E}_{as}$, the function $-\theta_{as,2}(-Y)$ satisfies the same BVP as $\theta_{as,2}$. Uniqueness gives $\theta_{as,2}(-Y) = -\theta_{as,2}(Y)$ and thus an even derivative. The inward-normal definition of the two wall fluxes then immediately yields $Nu_{2,0}(+1) = -Nu_{2,0}(-1)$.

S5.6 Proof of linearity in Grashof number

The thermal modes are independent of Gr, while the hydrodynamic BVP is linear with forcing proportional to $Gr\Theta'$. Consequently $\widehat{V}, \widehat{U}, \mathcal{E}_{as}$ and Π_T are each proportional to Gr at fixed (k, Re, Pr).

S5.7 Integral representation of the pumping coefficient

For $\theta'' = f(Y)$ with $\theta(\pm 1) = 0$,

$$\theta'(1) = \frac{1}{2}\int_{-1}^{1}(1+Y)f(Y)\, dY, \qquad \theta'(-1) = -\frac{1}{2}\int_{-1}^{1}(1-Y)f(Y)\, dY. \tag{S10}$$

For odd $f = Pe\mathcal{E}_{as}$,

$$\theta'(1) = \theta'(-1) = \frac{Pe}{2}\int_{-1}^{1} Y\mathcal{E}_{as}(Y)\, dY, \tag{S11}$$

and the inward-normal definition gives Eq. (41).

S5.8 Independent validation by initial-value shooting

For the thermal BVP, the unknown complex derivative at $Y = -1$ is adjusted to satisfy the prescribed value at $Y = +1$. For the fourth-order hydrodynamic BVP, the two complex initial derivatives $\widehat{V}''(-1)$ and $\widehat{V}'''(-1)$ are adjusted until $\widehat{V}(+1) = \widehat{V}'(+1) = 0$. The shooting solutions are compared against the adaptive collocation solutions, and Π_T is independently reconstructed from the first-order fields.

S6 Construction of the two independent mixed kernels

Let $\Theta^{(s)}, \Theta^{(a)}$ denote the unit-amplitude first-order thermal modes and $\widehat{U}^{(s)}, \widehat{V}^{(s)}, \widehat{U}^{(a)}, \widehat{V}^{(a)}$ the corresponding hydrodynamic modes. Their bilinear combinations fix the two real profiles that carry the rectified response. Define

$$Z_1 = \widehat{V}^{(a)}\big(\partial_Y \Theta^{(s)}\big)^* + \widehat{U}^{(a)}\big(-ik\Theta^{(s)}\big)^*, \qquad Z_2 = \widehat{V}^{(s)}\big(\partial_Y \Theta^{(a)}\big)^* + \widehat{U}^{(s)}\big(-ik\Theta^{(a)}\big)^*, \tag{S12}$$

and, from them, the two independent mixed kernels

$$\mathcal{K}_c = \tfrac{1}{2}\Re(Z_1 + Z_2), \qquad \mathcal{K}_q = \tfrac{1}{2}\,\mathrm{Im}(Z_2 - Z_1). \tag{S13}$$

The parity argument of Theorem 1 applies to *each* kernel separately: both $\mathcal{K}_c$ and $\mathcal{K}_q$ are odd in Y, so each produces an odd mean temperature correction and wall-flux corrections that are equal and opposite. The in-phase kernel $\mathcal{K}_c$ is the one already used in the amplitude-asymmetric configuration, for which $d = 0$; the quadrature kernel $\mathcal{K}_q$ is the additional degree of freedom that phase-shifted forcing excites. Substituting (S13) into the streamwise-averaged mixed forcing gives Eq. (52) of the main manuscript, where the two pumping coefficients Π_T and $\Pi_T^{\perp}$, the exact phase law and its equal-amplitude corollary are stated and proved.

S7 Physical realization of the symmetry-selected pumping mechanism

The exact analysis above identifies the symmetry class of the nonlinear wall-flux redistribution, but the corresponding physical transport pathway is most clearly understood from the resolved two-dimensional reference solution. We therefore consider the representative parameter set

$$(k, Pe, Re, Pr, Gr) = (0.8, 7, 10, 0.7, 100), \tag{S14}$$

which lies well inside the finite parameter domain investigated in the present study. The wavelength is neither in the asymptotic long-wave limit nor in the short-wave regime, while the buoyancy forcing is sufficiently strong for the secondary hydrodynamic response to be clearly resolved. This case is used throughout this section as a physical reference state.

A central point of the present interpretation is that the nonlinear pumping cannot be inferred from the total velocity field alone. The base Poiseuille flow remains dominant, whereas the buoyancy-induced perturbation is much smaller in amplitude but is responsible for the transverse transport that enters the quadratic thermal interaction. The reference solution therefore provides a direct physical realization of the analytical

sequence

$$\theta_1 \longrightarrow \partial_Y \theta_1 \longrightarrow (U_1, V_1) \longrightarrow \mathcal{E}_{as} \longrightarrow \Pi_T. \tag{S15}$$

S7.1 Two-dimensional thermal structure and buoyancy-induced secondary motion

Figure S11 shows the first-order temperature response for the reference configuration together with its transverse gradient. The temperature field exhibits the imposed streamwise periodicity, while the transverse gradient is strongly structured across the channel. This distinction is important because the buoyancy term in the linearized momentum equation is proportional to $\Theta'(Y)$, rather than to the temperature amplitude itself. The resolved temperature field therefore provides the spatial structure of the thermal forcing that is subsequently converted into a hydrodynamic response.

The corresponding hydrodynamic response is shown in Fig. S12. The total streamwise velocity remains close to the Poiseuille reference profile,

$$U_0(Y) = \frac{3}{2}(1 - Y^2), \tag{S16}$$

which confirms that the perturbation is a weak correction to the underlying forced flow. In the reference calculation,

$$\frac{\max |U_1|}{\max |U_0|} \simeq 6.35 \times 10^{-3}, \tag{S17}$$

whereas the transverse velocity is of the same order as the perturbation streamwise velocity and is therefore dynamically relevant even though both remain small compared with the base flow.

For this reason, the streamlines shown in the right panel of Fig. S12 are constructed from the perturbation field (U_1, V_1) rather than from the total velocity $(U_0 + U_1, V_1)$. The latter would be overwhelmingly dominated by the nearly parallel Poiseuille component and would obscure the secondary transport. The perturbation streamlines reveal a streamwise-periodic secondary circulation induced by buoyancy. These secondary motions provide the transverse advection required to convert the imposed thermal modulation into a nonzero quadratic mean response.

The reference solution therefore gives a useful physical interpretation of the analytical scaling

$$\widehat{V}^{(j)} \propto Gr. \tag{S18}$$

Buoyancy does not alter the first-order thermal modes at fixed (k, Pe); instead, it provides the hydrodynamic mediator through which the thermal asymmetry is converted into a transverse transport process. This is also consistent with the exact switch-off result $\Pi_T =$

0 at $Gr = 0$. The profile-level signature of that odd response, for a strongly asymmetric wall forcing, is shown in Fig. S15.

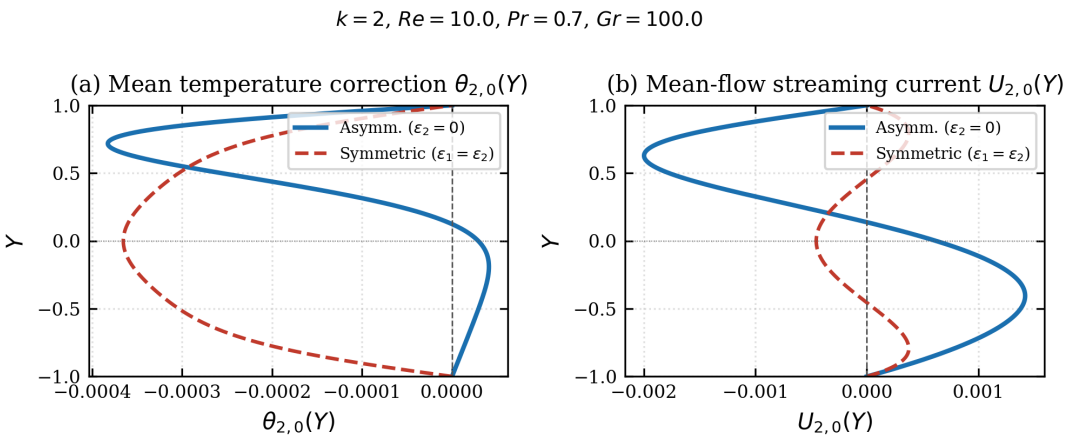


Figure S15: Cross-channel profiles of the mean temperature correction $\theta_{2,0}(Y)$ and the mean streaming velocity $U_{2,0}(Y)$ for a strongly asymmetric wall forcing. The odd thermal profile is the direct profile-level manifestation of the analytical symmetry theorem.

S7.2 Symmetry selection of the nonlinear interaction

The next step is to identify which part of the quadratic interaction is capable of producing a net wall-flux redistribution. Figure S13 shows the three quadratic forcing classes appearing in the streamwise-mean thermal equation. The self-interactions, $\mathcal{E}_{ss}$ and $\mathcal{E}_{aa}$, belong to the even symmetry class, whereas the mixed interaction $\mathcal{E}_{as}$ is odd under the reflection $Y \to -Y$.

The numerical structure shown in Fig. S13 provides a direct physical counterpart to the operator-based parity proof. The relevant observation is not simply that $\mathcal{E}_{as}$ changes sign across the channel, but that its positive and negative regions are organized in such a way that their unweighted integral cancels. Consequently, the mixed interaction does not act as a net volumetric heat source. Instead, it redistributes the mean thermal response transversely.

This distinction is essential. The nonlinear pumping observable is not controlled by the zeroth moment

$$\int_{-1}^{1} \mathcal{E}_{as}(Y)\, dY, \tag{S19}$$

which vanishes, but by its first transverse moment. From Eq. (41),

$$\Pi_T = -\frac{Pe}{2} \int_{-1}^{1} Y\, \mathcal{E}_{as}(Y)\, dY. \tag{S20}$$

The pumping mechanism can therefore be interpreted as a *transverse rectification* of

a locally balanced nonlinear interaction: the mixed forcing produces no net mean heat source, but its spatial distribution with respect to the channel centerline generates a finite redistribution of the two wall heat fluxes.

This interpretation also explains why the distinction between symmetry and magnitude must be maintained. The odd parity of $\mathcal{E}_{as}$ is fixed by reflection symmetry, whereas the magnitude and sign of its first moment remain determined by the coupled thermal–hydrodynamic dynamics.

S7.3 From local nonlinear interaction to wall-flux redistribution

The macroscopic consequence of the mixed interaction is the antisymmetric wall-flux correction established analytically above. For the fixed-Pe Reynolds-number comparison, Pr is adjusted according to $Pr = Pe/Re$; thus the cases $Re = 10$ and $Re = 35$ at $Pe = 7$ correspond to $Pr = 0.7$ and $Pr = 0.2$, respectively. For the present reference configuration ($Re = 10$, $Pr = 0.7$), the normalized pumping coefficient is

$$\Pi_T(Re = 10) = -4.7492 \times 10^{-2}, \tag{S21}$$

while at the same k and Pe, but with $Re = 35$ ($Pr = 0.2$),

$$\Pi_T(Re = 35) = 6.7256 \times 10^{-4}. \tag{S22}$$

Figure S14 displays the corresponding upper- and lower-wall corrections.

For $Re = 10$, the negative value of Π_T indicates that the upper-wall correction is directed opposite to the convention adopted for positive pumping, while the lower wall receives the compensating correction. The important point is that the total mean wall contribution remains balanced: the nonlinear mechanism redistributes the flux rather than creating an additional net heat input at second order.

The contrast between $Re = 10$ and $Re = 35$ is particularly informative. The first-order thermal problem depends on Re and Pr through the combination $Pe = RePr$, but the hydrodynamic response retains an explicit dependence on Re. Consequently, keeping Pe fixed does not preserve the nonlinear pumping response. The two cases therefore provide a direct physical demonstration of the parameter-dependence argument developed in Sec. 6: Reynolds number controls the hydrodynamic realization of the same thermal forcing.

S7.4 Hydrodynamic control of pumping direction

The sign reversal observed in Fig. S14 is the most useful physical consequence of the coupled formulation because it goes beyond the exact symmetry statement. At fixed $k = 0.8$, $Pe = 7$ and $Gr = 100$, the normalized pumping response changes sign between

$Re = 10$ and $Re = 35$, with a reversal near

$$Re_c \simeq 30.84. \tag{S23}$$

This behavior should not be interpreted as a change in the symmetry class. The relation

$$Nu_{2,0}(+1) = -Nu_{2,0}(-1) \tag{S24}$$

remains valid on both sides of the crossing. What changes is the sign of the first transverse moment in Eq. (S20). Increasing Re modifies the phase and spatial distribution of the buoyancy-induced transverse velocity relative to the thermal field, and therefore changes the signed overlap that enters $\mathcal{E}_{as}$. The reversal is thus a *dynamical* effect superimposed on an unchanged symmetry constraint.

In physical terms, the imposed thermal modulation establishes the spatial temperature pattern, buoyancy converts the transverse temperature gradient into a secondary circulation, and the Reynolds number controls how that circulation is phase-shifted and spatially weighted relative to the thermal pattern. The resulting quadratic overlap can consequently transfer the preferential wall-flux redistribution from one wall to the other without any change in the underlying reflection symmetry.

This provides a useful distinction between two classes of results obtained in the present study:

$$\boxed{\begin{array}{ll} \text{symmetry-controlled:} & Nu_{2,0}(+1) = -Nu_{2,0}(-1), \\ \text{dynamics-controlled:} & |\Pi_T|, \quad \mathrm{sign}(\Pi_T), \quad Re_c, \quad k^*(Pe). \end{array}} \tag{S25}$$

The first relation is exact under the assumptions of the theory, whereas the second set of quantities must be determined from the coupled thermal and hydrodynamic response.

S7.5 Physical picture of nonlinear thermal pumping

The resolved reference solution suggests a compact physical picture of the pumping mechanism. The periodic thermal boundary condition first creates a streamwise-organized temperature field. Its transverse gradient generates a buoyancy force, which produces a weak but spatially structured secondary velocity field. The interaction between the symmetric and antisymmetric components of this response generates the odd forcing $\mathcal{E}_{as}$. Although that forcing has zero net integral, its nonzero first transverse moment produces a finite wall-flux difference. The nonlinear pumping coefficient therefore measures a rectified transverse redistribution rather than a net creation of thermal energy.

The mechanism may be summarized schematically as

$$\text{periodic thermal pattern} \xrightarrow{\partial_Y \theta_1} \text{buoyancy-induced secondary flow} \xrightarrow{\text{quadratic interaction}} \mathcal{E}_{as}(Y) \xrightarrow{Y\text{-moment}} \Pi_T.$$

This interpretation also clarifies why the pumping disappears in the forced-convection limit $Gr = 0$. Without buoyancy, the secondary hydrodynamic mediator vanishes and the mixed thermal interaction cannot generate the wall-flux redistribution considered here. Conversely, when buoyancy is present, the symmetry-breaking boundary condition determines the admissible redistribution pattern, while the hydrodynamic parameters determine its magnitude and direction.

A useful physical interpretation follows from the geometry of the zero set. The disappearance of positive pumping for $Pe > Pe_c$ does not occur through a loss of the symmetry-breaking mechanism itself. The odd cross-interaction $\mathcal{E}_{as}$ remains structurally admissible, and the exact wall-flux antisymmetry remains unchanged. Instead, the region of parameter space in which the first transverse moment of $\mathcal{E}_{as}$ is positive contracts toward larger wavenumbers until the zero set reaches the upper boundary $k = 10$. The post-critical regime therefore corresponds to the expulsion of the sign-changing locus from the finite admissible domain, rather than to the disappearance of the underlying nonlinear interaction. This interpretation is consistent with the simultaneous preservation of the symmetry class and the continuous local sign change observed in $\Pi_T(10, Pe)$ near Pe_c.